

ProjectDAWIY：自分好みのDAW環境をDIYする共通プラットフォームの構築

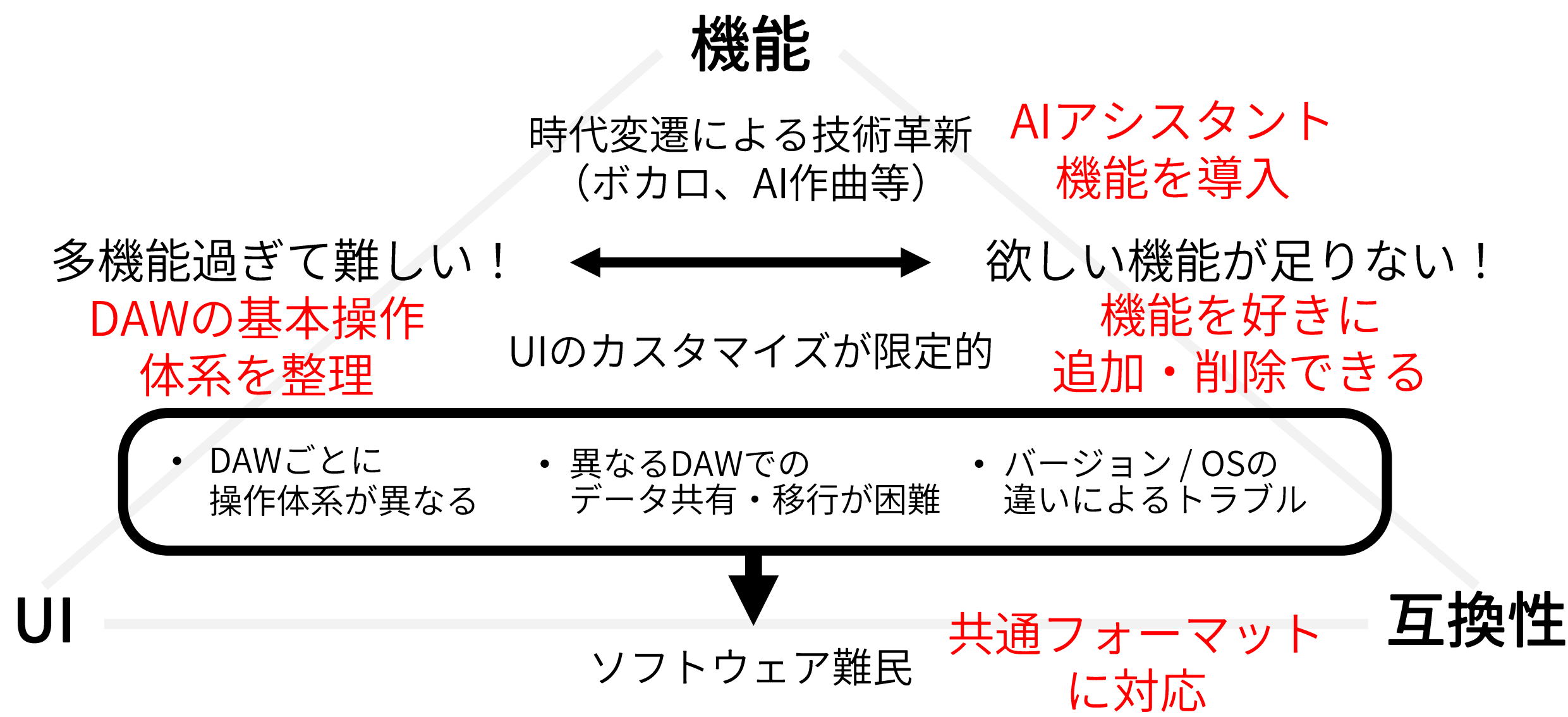
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GitHub

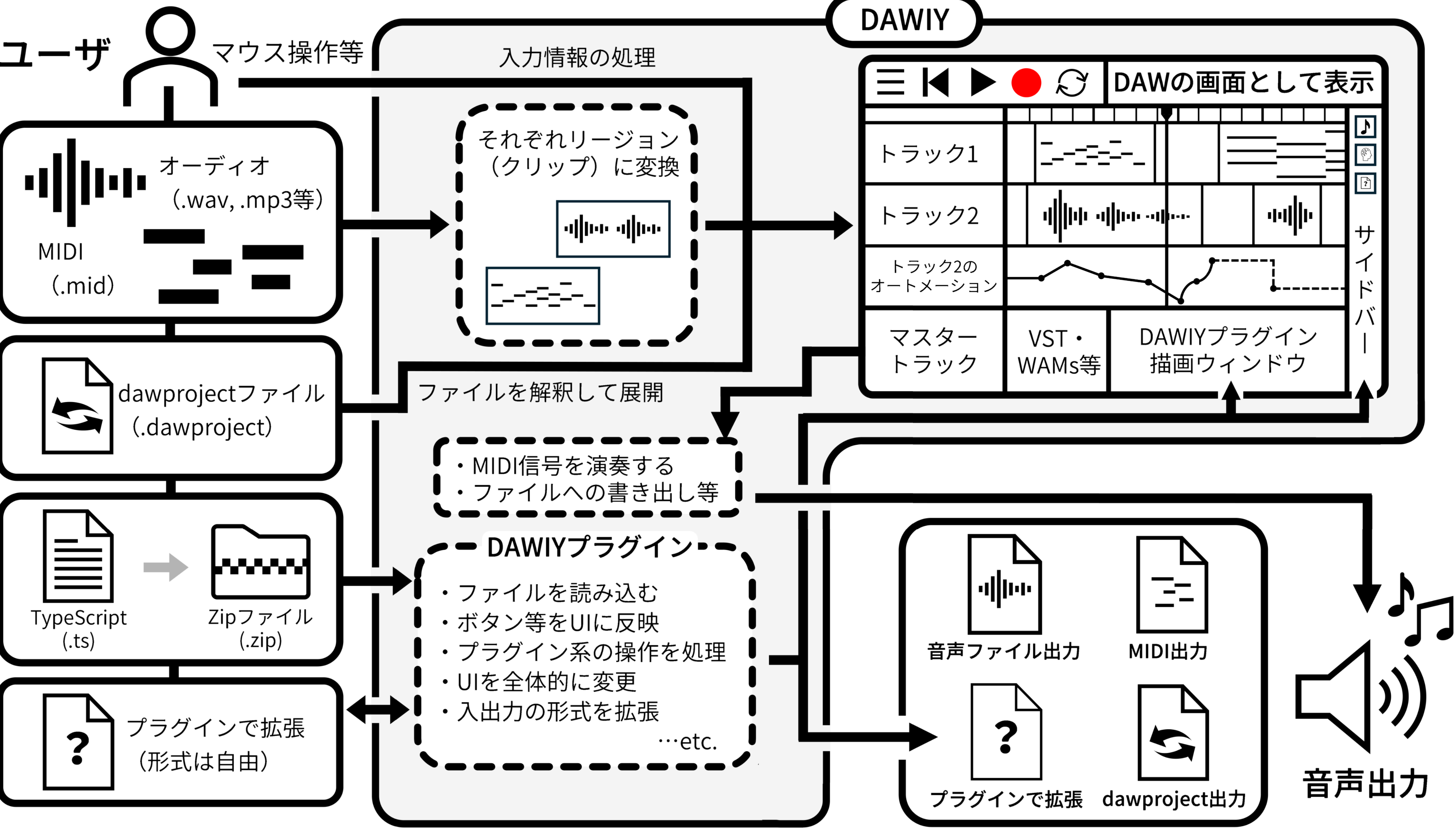
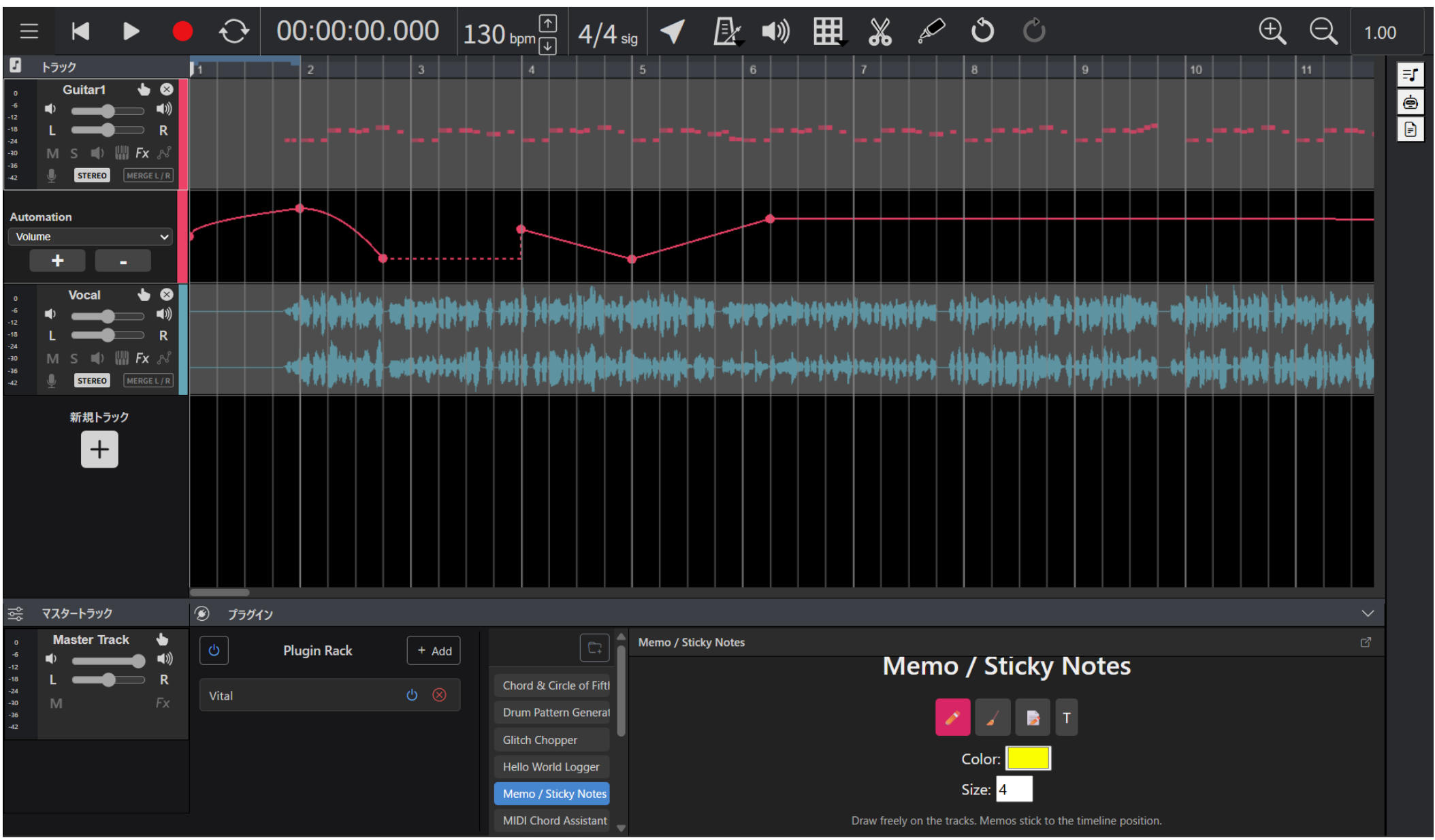
https://github.com/NumberZero-0621/DAWIY



DAWのかゆいところをDAWIYで解決



DAW+DIY=DAWIY



DAWIYプラグインとは…

VST プラグインではカバーできない、DAWの入出力・UI・操作等に介入する処理を拡張するための独自のプラグインシステム public/src/DawiyPluginsディレクトリ以下に配置された TypeScriptファイル (.ts) として管理

```
import App from "../App";
import { DawiyPlugin } from "../IDawiyPlugin";
import DawiyPluginBase from "../DawiyPluginBase";
import HostAPI from "../HostAPI";
import Track from "../Models/Track/Track";
import MIDIRegion from "../Models/Region/MIDIRegion";
import { MIDI, MIDINote } from "../Audio/MIDI/MIDI";
import { AUDIO_MIDIS_BPM, TEMPO } from "../Env";
```

```
@DawiyPlugin
export default class MelodyCreatorPlugin extends DawiyPluginBase {
  id = "melody-creator-plugin";
  name = "Melody Creator";
  description = "Add a simple 'Do-Re-Mi' sequence to a new track within a MIDI clip.";
  author = "AI Assistant";
  constructor(app: App) {
    super(app);
  }

  public override onInit(host: HostAPI) {
    host.ui.showToast("Melody Creator Plugin initialized!");
  }

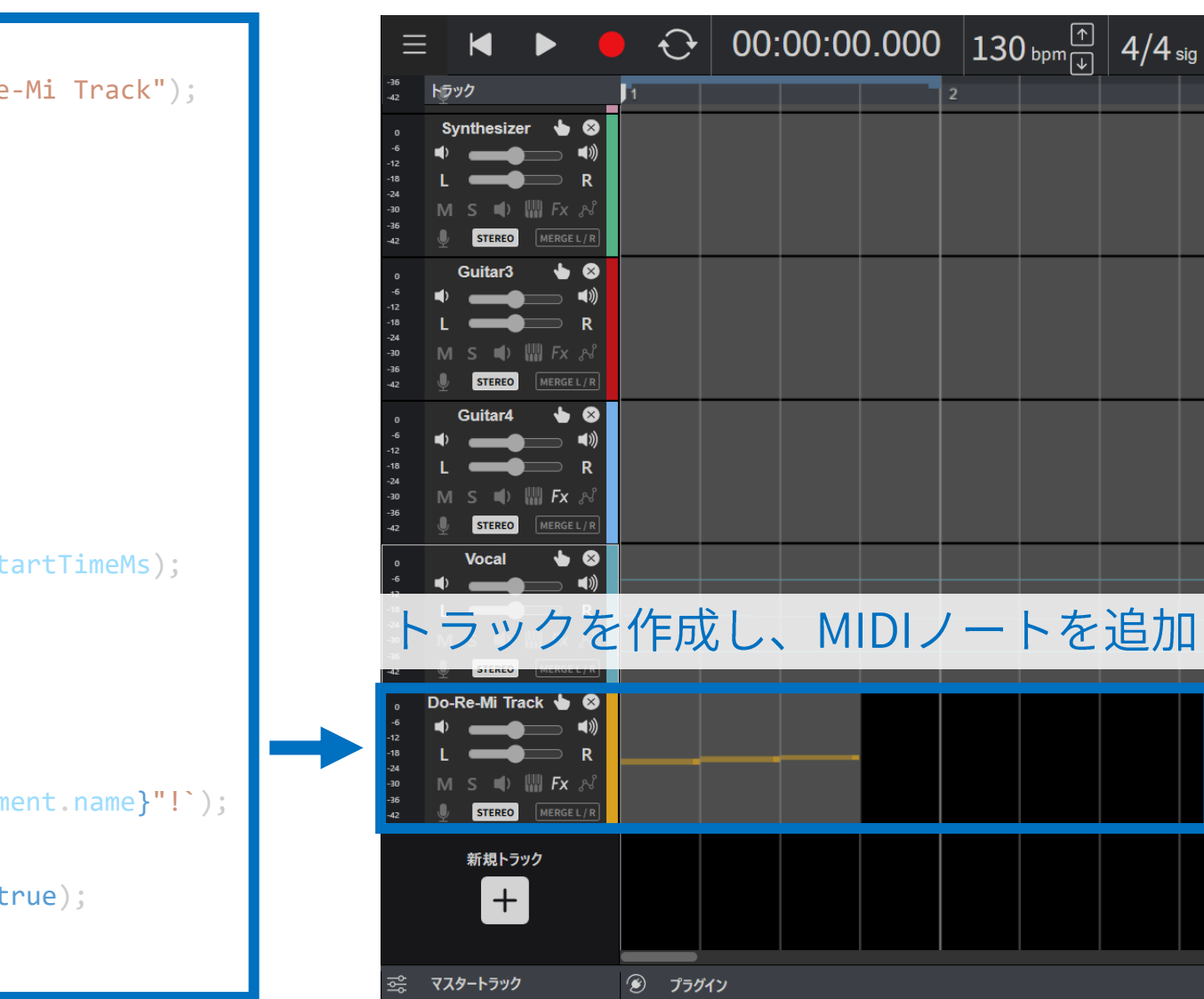
  public override render(container: HTMLElement) {
    container.className = "ap-3";
    container.innerHTML = `
      <h3>Melody Creator</h3>
      <div>Click the button below to add a new MIDI track with a simple C4-D4-E4 quarter-note melody starting from beat 0.</div>
      <button>Add Do-Re-Mi Track</button>
    `;
    const createMelodyButton = document.createElement("button");
    createMelodyButton.className = "btn btn-primary w-100 mt-3";
    createMelodyButton.textContent = "Add Do-Re-Mi Track";
    createMelodyButton.onclick = () => this.createDoReMiMelody();
    container.appendChild(createMelodyButton);
  }

  async createDoReMiMelody() {
    try {
      const newTrack = Track = await this.app.tracksController.createTrack("Do-Re-Mi Track");
      const notes = [
        { pitch: 60, name: "C4" },
        { pitch: 62, name: "D4" },
        { pitch: 64, name: "E4" },
      ];
      const beatsMs = (60 / TEMPO) * 1000;
      const durationMs = 3 * beatsMs;
      const mid = new MIDI(500, durationMs);
      const velocity = 100;
      for (let i = 0; i < notes.length; i++) {
        const note = notes[i];
        const startTimes = i * beatsMs;
        const noteDurationMs = beatsMs;
        mid.putNote(new MIDINote(note.pitch, velocity, 0, noteDurationMs, startTimes));
      }
      const startMs = 0;
      const newRegion = new MIDIRegion(mid, startMs);
      this.app.regionsController.addRegion(newTrack, newRegion);
      this.app.tracksController.select(newTrack);
      this.app.hostAPI.ui.showToast("Do-Re-Mi melody added to "${newTrack.name}"!");
    } catch (error: any) {
      host.ui.showToast("Error creating melody: ${error.message}", true);
      console.error("Error creating melody:", error);
    }
  }

  public override onDeactivate() {
  }
}
```

見た目の定義

ボタンを押すと発火



挙動の定義

